AN UPPER BOUND FOR THE DAVENPORT CONSTANT OF C_n^3

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Abstract. Let $\mathsf{D}(G)$ denote the Davenport constant of a finite abelian group G, and let C_n denote a cyclic group of order n. For $n \geq 2$, put

$$P(n) = \max_{p^a \| n} p^a,$$

the largest primary component of n. We prove the pointwise bound

$$\mathsf{D}(C_n^3) \leq 4n - P(n) - 2.$$

In particular, the normalized supremum

$$S := \sup_{n \geq 2} \frac{\mathsf{D}(C_n^3) - 1}{n - 1}$$

satisfies $S \leq 4$, improving the known uniform upper bound $S \leq 20369$ due to Zakarczemny [5]. The conjectural value remains $S = 3$. The proof combines known local zero-sum estimates over C_p^3 with an induction over the prime factors of $n/P(n)$, starting from the largest primary component of n. The proof was discovered with assistance from GPT-5.5 Pro. The deduction from the cited results in the literature was also checked in a conditional Lean 4 formalization with Aristotle [7].

1. Introduction and notation

Throughout, all finite abelian groups are written additively. For a finite abelian group G, the Davenport constant $\mathsf{D}(G)$ is the least integer N such that every sequence over G of length at least N contains a non-empty zero-sum subsequence. For $k \geq 1$, let $\mathsf{D}_k(G)$ be the least integer N such that every sequence over G of length at least N contains k pairwise disjoint non-empty zero-sum subsequences. Finally, $\eta(G)$ denotes the least integer N such that every sequence over G of length at least N contains a non-empty zero-sum subsequence of length at most $\exp(G)$.

We write C_n^3 for $C_n \oplus C_n \oplus C_n$. The main result is the following.

Theorem 1.1. *For every integer $n \geq 2$,*

$$\mathsf{D}(C_n^3) \leq 4n - P(n) - 2, \qquad P(n) = \max_{p^a \| n} p^a.$$

Consequently, $\mathsf{D}(C_n^3) \leq 4(n-1)$ for every $n \geq 2$.

In particular, if

$$S := \sup_{n \geq 2} \frac{\mathsf{D}(C_n^3) - 1}{n - 1},$$

then Theorem 1.1 gives $S \leq 4$. The known uniform upper bound is $S \leq 20369$, coming from the estimate

$$\mathsf{D}(C_n^3) \leq 20369(n-1) + 1$$

of Zakarczemny [5]. The standard lower bound gives $S \geq 3$ [4], and the conjectural pointwise formula is

$$\mathsf{D}(C_n^3) = 3(n-1) + 1 \qquad (n \geq 2),$$

which is equivalent to $S = 3$ [4]. Thus the result below improves the known interval from $3 \leq S \leq 20369$ to

$$3 \leq S \leq 4.$$

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The proof uses the inductive method for generalized Davenport constants, together with local estimates for $\mathsf{D}_k(C_p^3)$. We shall use the following standard facts from zero-sum theory. If

$$G \simeq C_{n_1} \oplus \cdots \oplus C_{n_r}, \qquad 1 < n_1 \mid \cdots \mid n_r,$$

then

$$\mathsf{D}(G) \geq \mathsf{D}^*(G) := 1 + \sum_{i=1}^{r}(n_i - 1),$$

and equality holds when G is a p-group; see Gao–Geroldinger [4]. Applying this to $C_{p^a}^3$ gives

$$\mathsf{D}(C_{p^a}^3) = 3p^a - 2.$$

The same lower bound applied to C_m^3 gives

$$\mathsf{D}(C_m^3) \geq 3m - 2.$$

Thus, in the sequel, we shall use

$$\mathsf{D}(C_{p^a}^3) = 3p^a - 2, \qquad \mathsf{D}(C_m^3) \geq 3m - 2. \tag{1}$$

2. Auxiliary estimates

We first record two consequences of the definitions.

Lemma 2.1 (Inductive method). *Let $H \leq G$ be finite abelian groups. Then*

$$\mathsf{D}(G) \leq \mathsf{D}_{\mathsf{D}(H)}(G/H).$$

Proof. Let S be a sequence over G of length at least $\mathsf{D}_{\mathsf{D}(H)}(G/H)$, and let $\overline{S}$ be its image in G/H. By the definition of $\mathsf{D}_{\mathsf{D}(H)}(G/H)$, the sequence $\overline{S}$ contains $\mathsf{D}(H)$ pairwise disjoint non-empty zero-sum subsequences. Lifting them to subsequences $T_1, \ldots, T_{\mathsf{D}(H)}$ of S, their sums lie in H. The sequence $\sigma(T_1), \ldots, \sigma(T_{\mathsf{D}(H)})$ over H has a non-empty zero-sum subsequence. The union of the corresponding T_i's is a non-empty zero-sum subsequence of S. □

Lemma 2.2 (Extraction lemma). *Let G be a finite abelian group, let $e = \exp(G)$, and let $j \geq 1$. Assume that $\eta(G) \leq M$ and $\mathsf{D}_j(G) \leq M$. Then, for every $k \geq j$,*

$$\mathsf{D}_k(G) \leq M + e(k - j).$$

Proof. Let S be a sequence over G of length at least $M + e(k-j)$. Repeatedly using $\eta(G) \leq M$, we may remove $k - j$ pairwise disjoint non-empty zero-sum subsequences, each of length at most e. The remaining sequence has length at least M, hence contains j further pairwise disjoint non-empty zero-sum subsequences. Altogether S contains k such subsequences. □

The following local estimate packages the known zero-sum results over C_p^3 that are needed in the proof.

Lemma 2.3 (Local estimate). *For every prime p and every $k \geq 3p - 2$,*

$$\mathsf{D}_k(C_p^3) \leq pk + p^2.$$

Proof. For $p = 2$, Freeze–Schmid record that $D_0(C_2^3) = 3$ and $k_D(C_2^3) = 2$ in their asymptotic notation [3]. Thus

$$\mathsf{D}_k(C_2^3) = 2k + 3 \qquad (k \geq 2),$$

and the desired estimate follows from $2k + 3 \leq 2k + 4$.

For $p = 3$, Bhowmik–Schlage-Puchta prove that every sequence of length 15 over C_3^3 contains three disjoint zero-sum subsequences [1]; this input is also cited in their later paper, where they use $\eta(C_3^3) = 17$ [2]. Hence $\mathsf{D}_3(C_3^3) \leq 17$. Applying Lemma 2.2 with $M = 17$ and $j = 3$, we get, for all $k \geq 3$,

$$\mathsf{D}_k(C_3^3) \leq 17 + 3(k-3) = 3k + 8 \leq 3k + 9.$$

For $p = 5$, the same paper gives $\eta(C_5^3) = 33$ and shows that every sequence of length 33 over C_5^3 contains three disjoint zero-sum subsequences [2]. Thus $\mathsf{D}_3(C_5^3) \le 33$, and Lemma 2.2 gives, for all $k \ge 3$,

$$\mathsf{D}_k(C_5^3) \le 33 + 5(k-3) = 5k + 18 \le 5k + 25.$$

It remains to consider $p \ge 7$. Put

$$M = (p+1)(3p-7) + 4 = 3p^2 - 4p - 3.$$

By Bhowmik–Schlage-Puchta [2],

(2) $$\eta(C_p^3) \le M,$$

and, for every $j \ge 1$,

(3) $$\mathsf{D}_j(C_p^3) \le \max\left\{5p-2, \frac{3(p-1)}{2}j + 2p + 5\right\}.$$

Set

$$x = \frac{M - 2p - 5}{\frac{3}{2}(p-1)} = \frac{2(3p^2 - 6p - 8)}{3(p-1)}, \qquad j_0 = \lfloor x \rfloor.$$

Since $p \ge 7$, we have $5p - 2 \le M$. Also, by the definition of j_0,

$$\frac{3(p-1)}{2}j_0 + 2p + 5 \le M.$$

Therefore (3) gives $\mathsf{D}_{j_0}(C_p^3) \le M$. Moreover $j_0 < 2p < 3p-2$. Finally, since $j_0 \ge x - 1$,

$$pj_0 + p^2 \ge p(x-1) + p^2 = M + \frac{3p^2 - 16p - 9}{3(p-1)} \ge M,$$

the last inequality again following from $p \ge 7$. Now let $k \ge 3p-2$. Then $k \ge j_0$, and Lemma 2.2, together with (2) and $\mathsf{D}_{j_0}(C_p^3) \le M$, gives

$$\mathsf{D}_k(C_p^3) \le M + p(k - j_0).$$

Using $M \le pj_0 + p^2$, we obtain

$$\mathsf{D}_k(C_p^3) \le pj_0 + p^2 + p(k - j_0) = pk + p^2.$$

This proves the lemma in all cases. □

3. Proof of the main theorem

Let $n \ge 2$, and put

$$Q = P(n).$$

By definition, Q is a prime-power divisor of n. Factor the remaining part of n as

$$n = Qp_1p_2\cdots p_s,$$

where $p_1, \dots, p_s$ are primes, counted with multiplicity. Since Q is the largest primary component of n, each $p_i \le Q$. Put

$$m_0 = Q, \qquad m_i = Qp_1\cdots p_i \quad (1 \le i \le s).$$

We prove by induction on i that

(4) $$\mathsf{D}(C_{m_i}^3) \le 4m_i - Q - 2.$$

For $i = 0$, the integer $m_0 = Q$ is a prime power. Hence (1) gives

$$\mathsf{D}(C_Q^3) = 3Q - 2 = 4Q - Q - 2,$$

which is (4) for $i = 0$.

Assume (4) for $m = m_{i-1}$, and write $p = p_i$. The group C_{pm}^3 contains a natural subgroup $H \simeq C_m^3$, with quotient $C_{pm}^3/H \simeq C_p^3$. By Lemma 2.1,

$$\mathsf{D}(C_{pm}^3) \le \mathsf{D}_{\mathsf{D}(C_m^3)}(C_p^3).$$

Furthermore, by (1),

$$\mathsf{D}(C_m^3) \geq 3m - 2 \geq 3p - 2,$$

since $m \geq Q \geq p$. Lemma 2.3, applied with $k = \mathsf{D}(C_m^3)$, yields

$$\mathsf{D}(C_{pm}^3) \leq p\mathsf{D}(C_m^3) + p^2.$$

Using the induction hypothesis,

$$\mathsf{D}(C_{pm}^3) \leq p(4m - Q - 2) + p^2.$$

It remains to compare this with $4pm - Q - 2$. The required inequality is

$$p^2 - 2p + 2 \leq (p - 1)Q.$$

Since $Q \geq p$,

$$(p - 1)Q \geq p(p - 1) = p^2 - p \geq p^2 - 2p + 2$$

for every prime $p \geq 2$. Therefore

$$\mathsf{D}(C_{pm}^3) \leq 4pm - Q - 2,$$

which proves the induction step.

Taking $i = s$, we have $m_s = n$, and hence

$$\mathsf{D}(C_n^3) \leq 4n - Q - 2 = 4n - P(n) - 2.$$

Since $P(n) \geq 2$, this implies

$$\mathsf{D}(C_n^3) \leq 4n - 4 = 4(n - 1).$$

This proves Theorem 1.1.

4. Conclusion

We have shown that

$$S = \sup_{n \geq 2} \frac{\mathsf{D}(C_n^3) - 1}{n - 1} \leq 4.$$

This improves the known uniform upper bound $S \leq 20369$ to $S \leq 4$. The conjectural value $S = 3$, equivalent to the pointwise formula

$$\mathsf{D}(C_n^3) = 3(n - 1) + 1 \qquad (n \geq 2),$$

remains open.

5. Remark on formal verification

A conditional Lean formalization of the deduction in this note is available in [6]. The formalization assumes, as external axioms, the zero-sum inputs cited above: the standard p-group formula and lower bound, the inductive method, and the local estimates over C_p^3. It checks the global induction and the resulting pointwise estimate of Theorem 1.1.

References

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